Restaurant Market Expansion Screener

Product Requirements Document (PRD)

Document control	
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Version	1.2 (supersedes Version 1.1; Version 1.0 dated 20 October 2026 is retained unchanged)
Date	1 November 2026
Project	Portfolio analytics project built in Google Sheets and Excel using the Zomato restaurant dataset
Evidence used	Project workbook (.xlsx export), README, solution document, analysis deck and dashboard screenshots. Figures marked "file check" were recalculated from the Cleaned Data sheet of the exported workbook. The live Google Sheet was not opened.

Revision history

Version	Date	Summary of changes
1.0	20 Oct 2026	Next-version specification. Requirements and acceptance criteria; existing features to be verified before being called complete.
1.1	1 Nov 2026	Rewritten to describe the actual project. Implementation status, metric definitions, UAT checklist and future enhancements added. Prepared before the project files were inspected, so most items were marked pending validation.
1.2	1 Nov 2026	Statuses updated after inspecting the workbook, README, solution document and dashboard. Nine-record difference between 9,551 and 9,542 traced to its cause and moved from open item to resolved, with documentation clean-up outstanding. Dashboard findings recorded: charts are pivot-backed; slicer behaviour not confirmed from the export; no price-range slicer; some text still shows pre-cleaning counts. UAT checklist completed for checks that could be run from the workbook file.

1. Executive Summary

The Restaurant Market Expansion Screener is a Google Sheets analysis of the Zomato restaurant dataset. It asks where a new restaurant brand should look first outside India. It compares countries and cities on restaurant listings, votes, ratings, cost for two and price range, and is meant to narrow the research scope. It is a screening tool, not an investment decision.

The original dataset has 9,551 restaurant records and a 15-row country reference table. The cleaned analysis dataset has 9,542 records. The nine-record difference has been traced: the nine records are all United States restaurants with a blank Cuisines field, and they were excluded from the Cleaned Data sheet (Section 7.1). India accounts for 8,652 of the 9,542 cleaned records (90.7%), so every other country has a much smaller sample.

The work covers data cleaning, a duplicate check, country lookup, derived fields (rating status, opening year, primary cuisine, cost in INR), country, city and cuisine summaries using pivot tables and COUNTIFS and AVERAGEIFS formulas, and a Google Sheets dashboard with six KPI cards, nine

charts, and Year and Country slicers. The analysis shortlists Indonesia, Turkey, Qatar and Sri Lanka for deeper research.

This version updates the PRD with what the project files show. Headline figures and the shortlist statistics were recalculated and match. Open items remain: several documents and the dashboard text still quote the pre-cleaning counts (9,551 and 7,403); slicer behaviour could not be confirmed from the exported file; exchange rates are fixed in a formula with no recorded date; and a Suggested_Flag column does not match the final shortlist. Section 8 lists them.

2. Background, Goals and Scope

2.1 Background

India is the largest market in the dataset. A company looking abroad needs a quick, consistent way to compare the other 14 countries before spending on detailed research. The restaurant data shows what is listed and how diners rate it, so it can support a first-stage comparison. It cannot show demand or profitability.

2.2 Goals

Compare the 15 countries, and selected cities, on restaurant count, rating, votes, cost for two and price range; use definitions that give the same answer when recalculated; and record data-quality decisions and limitations so results are read with the right caution.

2.3 Users

User	Need
Strategy or business analyst	Compare countries and cities using consistent definitions and filters.
Business stakeholder	See which markets deserve deeper research and understand the limits of the data.
Data owner or reviewer	Trace each figure to a cleaning rule or formula.

2.4 Scope

In scope	Out of scope
Cleaning and preparation of the Zomato data in a separate sheet Country lookup and country, city and cuisine summaries Pivot-table analysis of ratings, votes, cost and price range Dashboard with Year and Country slicers Documentation of method, findings and limitations	Live or scheduled data refresh Orders, revenue, profitability or demand forecasting Claims that a market will succeed External data such as population, competition, rent or operating costs

2.5 Project files

File	Content
Workbook (.xlsx, export of Google Sheet)	Raw Data, country description, Cleaned Data, 15 analysis sheets and the Dashboard. Slicers and some charts do not work in Excel; the live Google Sheet is the working version.
Solution document (.docx)	Question-by-question solution with cleaning log, formulas, tables and findings. The README lists this file as the PRD, but it is the solution document.

File	Content
Analysis deck (.pptx), README, chart images	Presentation of the analysis, repository overview and dashboard screenshots.

3. Implementation Status

"Verified" means the item was recalculated or inspected in the exported workbook for this revision. Items that depend on the live Google Sheet, or on a statement in the project documents that was not reproduced, are marked implemented but not fully verified.

Area	Status	Evidence and open point
Raw data preserved and loaded	Implemented and verified	Raw Data sheet holds 9,551 records with no duplicate RestaurantID. Cleaned Data is a separate sheet.
Country lookup (15 countries)	Implemented and verified	VLOOKUP exact match on country code. All 9,542 cleaned records have a country name; no unmatched codes.
Rating treatment (1.0 as not rated)	Implemented and verified	Rating_Clean is blank where Rating is 1.0. 2,148 not rated; 7,394 rated in Cleaned Data. The treatment is an author assumption (Section 7.2).
Zero coordinates and zero costs blanked	Implemented and verified	Counts agree with the cleaning log (497 rows with both coordinates zero; 18 zero costs in raw, 15 remaining after the nine exclusions).
Duplicate check	Implemented and verified	No duplicate RestaurantID and no fully duplicated rows. No rows were removed as duplicates.
Exclusion of nine records with blank Cuisines	Implemented and verified	Nine raw IDs absent from Cleaned Data; all nine are US records with blank Cuisines. Documentation to be aligned (Section 7.1).
Text and currency label corrections	Implemented, not fully verified	Listed in the cleaning log (double spaces, Philippines, UK and UAE currency labels). Not re-checked cell by cell.
Cost for two converted to INR	Implemented, not fully verified	Fixed exchange rates sit inside a nested IF formula. Rate source and date are not recorded. The author notes the Turkish conversion is still to be verified.
Country, city and cuisine summaries	Implemented and verified	Counts, rated sample, average rating and average votes by country recalculated and matched. Shortlist figures (share rated 4+, city averages, price and rating correlation) matched.
Dashboard KPI cards (6)	Implemented and verified	Stored values (9,542; 7,394; 3.44; 18.6%; 156.77; 22.5%) match recalculation. Dashboard text and screenshot still show 9,551 and 7,403.
Dashboard charts (9)	Implemented, not fully verified	All nine reference pivot-table ranges. Rendering and filtering were not tested in the live sheet. Some country labels are truncated in the screenshot.

Area	Status	Evidence and open point
Year and Country slicers	Implemented, not fully verified	Both slicers exist. The dashboard note says both were tested. The export does not preserve slicer-to-pivot connections, so this could not be confirmed.
Price-range filter	Not implemented	Price range appears as a chart only. There is no price-range slicer.
Methodology and limitations documentation	Partially implemented	Cleaning log, README limitations and dashboard notes exist. Record counts are inconsistent between documents.
Weighted screening score, automated refresh, advanced dashboard features	Proposed future enhancement	Not built (Section 10). Suggested_Flag in Cleaned Data is not a weighted score.

4. Data and Methodology

4.1 Data sources

Source	Description
Restaurant dataset	Zomato restaurant records: 9,551 in Raw Data (22 columns), 9,542 in Cleaned Data (33 columns).
Country reference table	15 rows mapping country code to country name, used in the lookup.

4.2 Data preparation

The raw data was kept unchanged and all transformations were made in the Cleaned Data sheet. The steps below are taken from the workbook and the cleaning log.

Step	Method	Check
Rating placeholder	Rating of 1.0 treated as not rated. Rating_Clean = IF(Rating=1,"",Rating); Rating_Status = Rated or Not rated.	Verified
Zero values	Zero longitude, latitude and cost for two blanked in new or corrected columns.	Counts agree with log
Duplicates	RestaurantID checked for repeats. None found.	Verified
Missing cuisines	Nine records with blank Cuisines excluded from Cleaned Data.	Verified
Inconsistent text and labels	Extra spaces removed; Philippines, UK and UAE currency labels corrected (TRIM, IF, find and replace).	Per log; not re-checked
Country name	VLOOKUP(CountryCode, country table, 2, FALSE).	Verified
Derived fields	Opening_Date and Opening_Year (YEAR), Rated_4plus, Primary_Cuisine (first listed cuisine), Cost in INR.	Rated_4plus and year verified; cost not

4.3 Formulas and pivot tables

Country, city and cuisine summaries use COUNTIFS, AVERAGEIFS, MINIFS and MAXIFS on the Cleaned Data columns, with CORREL and FILTER used in the rating tests. The dashboard has ten pivot tables, all with source range Cleaned Data A1:AG9543, so they cover the full 9,542 records. The KPI cards read from a pivot table, so they respond to a slicer only if that pivot is connected to it.

5. Analytical Requirements

Definitions apply to every table, pivot table and KPI. The last column shows the value recalculated from Cleaned Data for all 15 countries combined (9,542 records).

Metric	Definition and calculation	Notes	Recalculated value
Restaurant count	Count of RestaurantID in the group. RestaurantID is unique, so records equal restaurants.	Based on cleaned data (nine exclusions).	9,542
Rated sample size	Count of non-blank Rating_Clean in the group.	Always read with the average rating.	7,394
Average rating	Mean of Rating_Clean. Not-rated records excluded.	Rating 1.0 treated as not rated.	3.44
Share rated 4+	Records with Rating_Clean of at least 4.0 divided by rated records.	Denominator is rated records, not all records.	18.6%
Votes	Mean Votes per record in the group, including not-rated records.	Shows engagement, not market share.	156.77
Not rated share	Not-rated records divided by all records.		22.5%
Average cost for two	Mean of Average_Cost_for_two (zeros blanked), converted to INR with fixed rates.	Depends on the exchange-rate assumptions.	Not recalculated
Price-range distribution	Records in Price_range 1 to 4 divided by all records in the group.	Shares sum to 100%.	46.5%, 32.6%, 14.7%, 6.1%
Country comparison	Table of the metrics above for each country.	Reference values in Appendix B.	See Appendix B

The shortlist figures in the README were also recalculated and match: share rated 4+ of Turkey 88.2%, Indonesia 81.0%, Sri Lanka 65.0% and Qatar 55.0%; city averages Jakarta 4.36, Ankara 4.31, Istanbul 4.29, Doha 4.06 and Colombo 3.87; price range and rating correlation about 0.02 for ranges 2 to 4.

6. Dashboard Requirements

The dashboard has six KPI cards, nine charts and two slicers. The table shows each requirement against what the workbook contains. Earlier feedback on charts, slicer connections and pivot backing is addressed where the workbook shows it, and the remaining gaps are listed.

ID	Requirement	Status	Evidence and outstanding work
DB-01	KPI cards for restaurants, rated restaurants, average rating, share rated 4+, average votes and not-rated share.	Implemented and verified	Values match recalculation. Dashboard heading and notes still say 9,551 and 7,403.
DB-02	Country charts: restaurants by country, average rating by country, average cost in INR.	Implemented, not fully verified	Pivot-backed. Cost chart depends on undocumented exchange rates. Country labels are truncated in the screenshot.
DB-03	Price-range distribution chart.	Implemented and verified	Pivot-backed. Shares match recalculation.
DB-04	Year slicer updating KPI cards, pivots and charts.	Implemented, not fully verified	Author states it was tested. Connection to each pivot needs confirmation in the live sheet.
DB-05	Country slicer updating KPI cards, pivots and charts.	Implemented, not fully verified	As DB-04.
DB-06	Price-range slicer.	Not implemented	Optional. Not in the current dashboard.
DB-07	Every chart backed by a pivot table on Cleaned Data.	Implemented and verified	All nine charts reference pivot ranges on the Dashboard sheet. All ten pivots use A1:AG9543.
DB-08	Dashboard notes consistent with the cleaned data.	Outstanding	Heading, methodology note and key insights quote 9,551 records and 7,403 rated.
DB-09	No unused or truncated formula blocks on the dashboard sheet.	Outstanding	Rows 132 to 151 hold helper formulas using ranges that stop at row 682. No chart references them. Remove or correct.
DB-10	Readable fallback for missing country codes.	Not implemented	VLOOKUP has no IFNA wrapper. No unmatched codes exist today, so the risk is low. Add before any data refresh.

7. Data Quality, Reconciliation and Limitations

7.1 Reconciliation of 9,551 and 9,542 records

Status: resolved. Comparing RestaurantID between Raw Data and Cleaned Data shows nine IDs in the raw data that are not in the cleaned data, and no IDs in the cleaned data that are not in the raw data. All nine are United States restaurants with a blank Cuisines field, and they are the only nine blank-Cuisines records in the raw data. The cleaning log lists "Missing Cuisines: 9" as excluded.

RestaurantID	Restaurant	City
17606621	HI Lite Bar & Lounge	Miller
17284211	Pearly's Famous Country Cookng	Albany
17284105	Cookie Shoppe	Albany
17374552	Corkscrew Cafe	Gainesville
17501439	Dovetail	Macon
17059060	Hillstone	Orlando

RestaurantID	Restaurant	City
17284158	Jimmie's Hot Dogs	Albany
17142698	Leonard's Bakery	Rest of Hawaii
17616465	Tybee Island Social Club	Savannah

Effect of the exclusion (not-rated records are 2,148 in both):

Measure	Raw (9,551)	Cleaned (9,542)
United States records	434	425
Rated records	7,403	7,394
Average rating	3.44	3.44
Average votes	156.91	156.77

No shortlisted country is affected, because all nine records are in the United States. The exclusion was an analyst decision made because cuisine analysis needs a cuisine value. An alternative would have been to keep the records with an "Unknown" cuisine. Three of the nine also had a zero cost for two.

Documentation still to align: the dashboard heading, methodology note and key insights, the screenshot, the README and the solution document (which says "9,551 rows kept" and "All 9,551 records" in the cleaning section) should state 9,551 original records and 9,542 cleaned records, and give the reason for the difference.

7.2 Known data-quality issues

Issue	Detail
Rating of 1.0 as not rated	It is the lowest rating in the file and 2,148 records have it, 2,139 of them in India. Treating it as a placeholder is a reasonable but unproven assumption.
Small samples outside India	Cleaned counts range from 4 (Canada) to 425 (United States). City samples are 14 to 20 listings.
Exchange rates	Fixed inside a nested IF formula with no source or date. Turkish conversion flagged by the author as still to be verified.
Suggested_Flag column	Flags Canada, Singapore, Qatar, Philippines and Indonesia (87 rows). The final shortlist is Indonesia, Turkey, Qatar and Sri Lanka.
Corrupted characters	Some names, addresses and localities still show encoding errors. The author notes this does not affect counts or averages.
Stray formulas	Q-4 I Subjective column L holds formulas with shifted references and a misspelt 'Qator'. They are not used by the dashboard.

7.3 Limitations

Restaurant listings and ratings alone cannot establish market demand, revenue, profitability or expansion feasibility. The dataset has no sales, footfall, rent or operating-cost data. Votes show recorded engagement, not market share. Rating behaviour may differ between countries, and cuisine results are associations in small samples. The output narrows where to research next; it does not say where to expand.

8. Outstanding Requirements

No.	Requirement	Reference
1	Update dashboard heading, notes, screenshot, README and solution document to show 9,551 original and 9,542 cleaned records, with the reason.	DB-08, Section 7.1
2	Confirm in the live Google Sheet that each slicer updates every KPI card, pivot and chart, and record the result.	DB-04, DB-05
3	Align Suggested_Flag with the final four-country shortlist, or rename or remove it.	Section 7.2
4	Record exchange rates with source and date, and verify the Turkish conversion.	DB-02
5	Remove or correct the helper formula blocks on the dashboard sheet and the stray formulas in Q-4 I Subjective.	DB-09
6	Fix truncated country labels on the bar charts.	DB-02
7	Replace the PRD link in the README with the actual PRD, and state the file versions.	Section 2.5

9. Acceptance Criteria and UAT

9.1 Acceptance criteria

ID	Criterion	UAT checks	Status
AC-01	Dashboard totals equal the cleaned dataset and the difference from the raw data is explained record by record.	4, Sect. 7.1	Met
AC-02	All 15 countries appear in the country summary.	5	Met
AC-03	Counts, rated sample, average rating, votes and price-range shares match independent calculation.	4 to 7	Met
AC-04	Average cost for two in INR can be reproduced from documented exchange rates.	8	Not met
AC-05	Every filter updates all connected KPI cards, pivots and charts.	9 to 12	Partly met
AC-06	No formula error values in any sheet.	2	Met
AC-07	Every chart is pivot-backed and readable.	13, 14	Partly met
AC-08	Record counts are consistent across the dashboard, README, deck and solution document.	15	Not met
AC-09	Cleaning steps, definitions and limitations are documented.	16, 17	Met
AC-10	Shortlist labels are consistent across the workbook and documents.	18	Not met

9.2 UAT checklist

"File check" means recalculated or inspected in the exported workbook. Checks that need the live Google Sheet remain for the author to run and record.

No.	Area	Check	Result
1	Formulas	Country lookup returns a name for every record (0 unmatched of 9,542).	Pass (file check)
2	Formulas	No #N/A, #REF!, #VALUE! or similar values in any sheet.	Pass (file check)
3	Formulas	Rating_Clean is blank for exactly the 2,148 records with Rating 1.0.	Pass (file check)
4	Metrics	KPI values match recalculation: 9,542; 7,394; 3.44; 18.6%; 156.77; 22.5%.	Pass (file check)
5	Country summary	All 15 countries present; counts sum to 9,542; count, rated sample, average rating and votes match recalculation.	Pass (file check)
6	Metrics	Price-range shares sum to 100% and match the dashboard chart.	Pass (file check)
7	Metrics	Shortlist figures in the README match recalculation.	Pass (file check)
8	Metrics	Average cost for two in INR matches calculation with documented rates.	Not independently verified
9	Filters	Year slicer updates all KPI cards, pivots and charts.	Not independently verified
10	Filters	Country slicer updates all KPI cards, pivots and charts.	Not independently verified
11	Filters	Year and Country together match filtered source data.	Not independently verified
12	Filters	Price-range filter updates the dashboard.	Outstanding
13	Charts	All nine charts reference pivot-table ranges on Cleaned Data.	Pass (file check)
14	Charts	Titles, axes and country labels are readable.	Outstanding
15	Documentation	Record counts are consistent across all project documents.	Outstanding
16	Documentation	Cleaning log lists each issue, count and treatment.	Pass (document check)
17	Documentation	Limitations are stated in the README and the dashboard notes.	Pass (document check)
18	Documentation	Suggested_Flag agrees with the final shortlist.	Outstanding

10. Future Enhancements

These are not part of the current project and remain proposed until built and verified.

- **Weighted screening score.** A normalised score combining market size, rating quality and price fit, with component scores and sample sizes shown. The weights in Version 1.0 were illustrations and are not validated. A minimum rated-sample threshold would be needed.
- **Price-range slicer** and a country comparison view that handles several selected countries.
- **Exchange-rate table** with source and date, replacing the rates inside the formula, and an IFNA fallback on lookups.
- **Automated refresh** of the source data.
- **External indicators** such as population, competition and rent, if a source is identified.

Appendix A. Portfolio Description

Suggested wording, valid while the statuses above hold:

“Analysed 9,551 Zomato restaurant records across 15 countries in Excel and Google Sheets. Cleaned the data, traced a nine-record difference to restaurants with missing cuisine information, built country, city and cuisine summaries with pivot tables and formulas, and created a dashboard with KPI cards, nine charts and Year and Country slicers. Shortlisted four countries for deeper research and documented the data-quality limitations.”

Appendix B. Country Reference Values

Recalculated from Cleaned Data (9,542 records). Rated sample is the count of non-blank Rating_Clean. Average votes is the mean over all records in the country. Use these values for spot checks.

Country	Records	Rated	Avg rating	Avg votes
India	8,652	6,513	3.35	137.21
United States of America	425	422	4.04	430.88
United Kingdom	80	79	4.14	205.49
United Arab Emirates	60	60	4.23	493.52
South Africa	60	60	4.21	315.17
Brazil	60	55	4.11	19.62
New Zealand	40	40	4.26	243.02
Turkey	34	34	4.30	431.47
Australia	24	24	3.66	111.42
Philippines	22	22	4.47	407.41
Indonesia	21	21	4.30	772.10
Qatar	20	20	4.06	163.80
Sri Lanka	20	20	3.87	146.45
Singapore	20	20	3.58	31.90
Canada	4	4	3.58	103.00